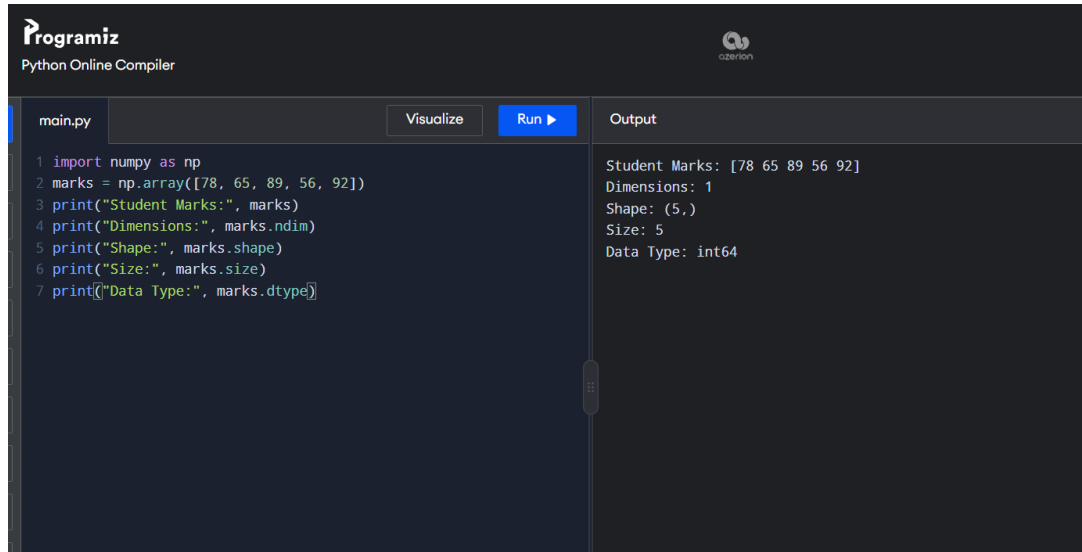


NUMPY ASSIGNMENT

Question 1 - Student Marks Array

The marks obtained by five students in a subject are given as [78, 65, 89, 56, 92]. Create a NumPy array and display the array along with its basic properties.



The screenshot shows the Programiz Python Online Compiler interface. The code editor on the left contains the following Python code:

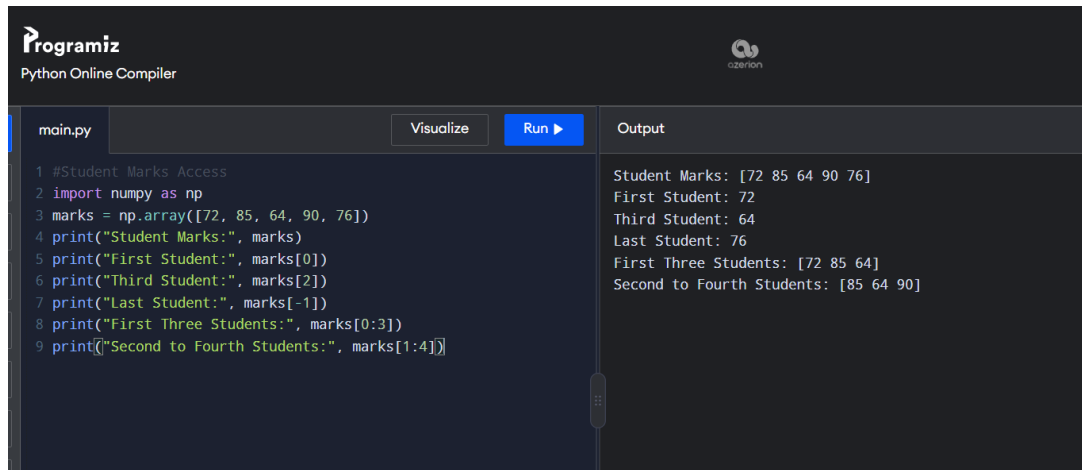
```
1 import numpy as np
2 marks = np.array([78, 65, 89, 56, 92])
3 print("Student Marks:", marks)
4 print("Dimensions:", marks.ndim)
5 print("Shape:", marks.shape)
6 print("Size:", marks.size)
7 print("Data Type:", marks.dtype)
```

The output panel on the right displays the results of the code execution:

```
Student Marks: [78 65 89 56 92]
Dimensions: 1
Shape: (5,)
Size: 5
Data Type: int64
```

Question 2- Student Marks Access

The marks of five students are stored in a NumPy array as [72, 85, 64, 90, 76]. Write a program to access and display specific student marks using NumPy indexing and slicing.



The screenshot shows the Programiz Python Online Compiler interface. The code editor on the left contains the following Python code:

```
1 #Student Marks Access
2 import numpy as np
3 marks = np.array([72, 85, 64, 90, 76])
4 print("Student Marks:", marks)
5 print("First Student:", marks[0])
6 print("Third Student:", marks[2])
7 print("Last Student:", marks[-1])
8 print("First Three Students:", marks[0:3])
9 print("Second to Fourth Students:", marks[1:4])
```

The output panel on the right displays the results of the code execution:

```
Student Marks: [72 85 64 90 76]
First Student: 72
Third Student: 64
Last Student: 76
First Three Students: [72 85 64]
Second to Fourth Students: [85 64 90]
```

Question 3- Subject-wise Marks

The marks obtained by five students in three subjects are given below. Create a NumPy array to represent the data and reshape it into an appropriate matrix format.[78, 85, 90, 65, 72, 80, 88, 91, 84, 56, 62, 70, 95, 89, 92]

ProgramizPython Online Compiler

main.py

Visualize

Run ▶

Output

⌕

⋮

```
1 #3
2 import numpy as np
3 marks = np.array([78, 85, 90, 65, 72, 80,
4                  88, 91, 84, 56, 62, 70,
5                  95, 89, 92])
6 matrix = marks.reshape(5, 3)
7 print("Original Array:", marks)
8 print("Reshaped Matrix:")
9 print(matrix)
10 print(f"Shape:", matrix.shape)
```

Original Array: [78 85 90 65 72 80 88 91 84 56 62 70 95 89 92]
Reshaped Matrix:
[[78 85 90]
 [65 72 80]
 [88 91 84]
 [56 62 70]
 [95 89 92]]
Shape: (5, 3)

Question 4- Internal and External Marks

The internal and external examination marks of five students are stored in two NumPy arrays. Write a program to calculate the final marks of each student using NumPy array operations.

ProgramizPython Online Compiler

main.py

Visualize

Run ▶

Output

⌕

⋮

```
1 #4
2 import numpy as np
3 internal = np.array([25, 28, 22, 30, 27])
4 external = np.array([55, 60, 48, 65, 58])
5 final = internal + external
6 print("Internal Marks:", internal)
7 print("External Marks:", external)
8 print(f"Final Marks:", final)
```

Internal Marks: [25 28 22 30 27]
External Marks: [55 60 48 65 58]
Final Marks: [80 88 70 95 85]

Question 5- Pass Percentage Analysis

The marks obtained by five students are [45, 78, 56, 32, 91]. Using NumPy Boolean masking, identify the students who have secured 50 marks or above.

Programiz

Python Online Compiler

main.py

Visualize

Run ▶

```
1 import numpy as np
2 marks = np.array([45, 78, 56, 32, 91])
3 passed = marks >= 50
4 print("Student Marks:", marks)
5 print("Pass Status:", passed)
6 print("Passed Students Marks:", marks[passed])
7 print("Number of Passed Students:", np.sum(passed))
8 print("Pass Percentage:", np.mean(passed) * 100)
```

Output

Student Marks: [45 78 56 32 91]
Pass Status: [False True True False True]
Passed Students Marks: [78 56 91]
Number of Passed Students: 3
Pass Percentage: 60.0

Question 6 - Average Marks

The marks of five students in three subjects are represented using a NumPy matrix. Write a program to calculate the average marks of each student.

Programiz

Python Online Compiler

main.py

Visualize

Run ▶

```
1 #6
2 import numpy as np
3 marks = np.array([
4     [78, 85, 90],
5     [65, 72, 80],
6     [88, 91, 84],
7     [56, 62, 70],
8     [95, 89, 92]
9 ])
10 average = np.mean(marks, axis=1)
11 print("Student Marks:")
12 print(marks)
13 print("Average Marks:")
14 print(average)
```

Output

Student Marks:
[[78 85 90]
 [65 72 80]
 [88 91 84]
 [56 62 70]
 [95 89 92]]
Average Marks:
[84.33333333 72.33333333 87.66666667 62.66666667 92.]

Question 7 - Class Performance Statistics

The marks obtained by five students are [67, 82, 91, 74, 58]. Using NumPy statistical functions, determine the total, average, highest, lowest, and standard deviation of the marks.

Programiz

Python Online Compiler

main.py

Visualize

Run ▶

```
1 #7
2 import numpy as np
3 marks = np.array([67, 82, 91, 74, 58])
4 print("Student Marks:", marks)
5 print("Total Marks:", np.sum(marks))
6 print("Average Marks:", np.mean(marks))
7 print("Highest Marks:", np.max(marks))
8 print("Lowest Marks:", np.min(marks))
9 print("Standard Deviation:", round(np.std(marks), 2))
```

Output

Student Marks: [67 82 91 74 58]
Total Marks: 372
Average Marks: 74.4
Highest Marks: 91
Lowest Marks: 58
Standard Deviation: 11.46

Question 8 - Subject-wise Performance

The marks of five students in three subjects are stored in a NumPy matrix. Write a program to calculate the total marks obtained in each subject using an appropriate axis operation.

Programiz

Python Online Compiler

main.py

Visualize

Run ▶

```
1 #8
2 import numpy as np
3 marks = np.array([
4     [78, 85, 90],
5     [65, 72, 80],
6     [88, 91, 84],
7     [56, 62, 70],
8     [95, 89, 92]
9 ])
10 total = np.sum(marks, axis=0)
11 print("Student Marks:")
12 print(marks)
13 print("Subject-wise Total:", total)
```

Output

Student Marks:
[[78 85 90]
 [65 72 80]
 [88 91 84]
 [56 62 70]
 [95 89 92]]
Subject-wise Total: [382 399 416]

Question 9 - Student-wise Performance

The marks of five students in three subjects are stored in a NumPy matrix. Write a program to calculate the total marks obtained by each student using an appropriate axis operation.

Programiz
Python Online Compiler

main.py

Visualize

Run ▶

```
1 #9
2 import numpy as np
3 marks = np.array([
4     [78, 85, 90],
5     [65, 72, 80],
6     [88, 91, 84],
7     [56, 62, 70],
8     [95, 89, 92]
9 ])
10 total = np.sum(marks, axis=1)
11 print("Student Marks:")
12 print(marks)
13 print("Student-wise Total:", total)
```

Output

Student Marks:
[[78 85 90]
 [65 72 80]
 [88 91 84]
 [56 62 70]
 [95 89 92]]
Student-wise Total: [253 217 263 188 276]

Question 10 - Student Ranking

The total marks obtained by five students are [245, 278, 219, 290, 256]. Use NumPy sorting and indexing operations to arrange the marks in order and determine the ranking of the students.

Programiz
Python Online Compiler

main.py

Visualize

Run ▶

```
1 import numpy as np
2 marks = np.array([245, 278, 219, 290, 256])
3 sorted_marks = np.sort(marks)
4 ranking = np.argsort(marks)[::-1]
5 print("Original Marks:", marks)
6 print("Ascending Order:", sorted_marks)
7 print("Descending Order:", sorted_marks[::-1])
8 print("Student Ranking:")
9 for i in range(len(ranking)):
10     print("Rank", i + 1, "- Student",
11           ranking[i] + 1, ":", marks[ranking[i]])
```

Output

Original Marks: [245 278 219 290 256]
Ascending Order: [219 245 256 278 290]
Descending Order: [290 278 256 245 219]
Student Ranking:
Rank 1 - Student 4 : 290
Rank 2 - Student 2 : 278
Rank 3 - Student 5 : 256
Rank 4 - Student 1 : 245
Rank 5 - Student 3 : 219

Question 11 -Duplicate Marks Analysis

The marks obtained by five students are [85, 92, 85, 76, 92]. Use NumPy functions to identify the unique marks obtained by the students.

Programiz

Python Online Compiler

main.py

Visualize

Run ▶

```
1 #11
2 import numpy as np
3 marks = np.array([85, 92, 85, 76, 92])
4 unique_marks = np.unique(marks)
5 print("Original Marks:", marks)
6 print("Unique Marks:", unique_marks)
7 print("Number of Unique Marks:", len(unique_marks))
```

Output

Original Marks: [85 92 85 76 92]
Unique Marks: [76 85 92]
Number of Unique Marks: 3

Question 12 - Missing Marks

The marks of five students are represented as [78, 85, np.nan, 92, 67], where np.nan represents a missing mark. Write a NumPy program to calculate the average marks without considering the missing value.

Programiz

Python Online Compiler

main.py

Visualize

Run ▶

```
1 #12
2 import numpy as np
3 marks = np.array([78, 85, np.nan, 92, 67])
4 average = np.nanmean(marks)
5 print("Student Marks:", marks)
6 print("Average Marks:", average)
7
```

Output

Student Marks: [78. 85. nan 92. 67.]
Average Marks: 80.5

Question 13 - Grade Classification

The marks obtained by five students are [95, 82, 74, 61, 45]. Using NumPy conditional operations, classify the students into appropriate grade categories based on their marks.

Programiz

Python Online Compiler

main.py

Visualize

Run ▶

```
1 #13
2 import numpy as np
3 marks = np.array([95, 82, 74, 61, 45])
4 grades = np.select(
5     [
6         marks >= 90,
7         marks >= 80,
8         marks >= 70,
9         marks >= 60
10    ],
11    ["A", "B", "C", "D"],
12    default="F"
13 )
14 print("Student Marks:", marks)
15 print("Grades:", grades)
```

Output

Student Marks: [95 82 74 61 45]
Grades: ['A' 'B' 'C' 'D' 'F']

Question 14 - Random Marks Generation

Generate marks for five students using NumPy's random number generation functionality.

Perform basic statistical analysis on the generated marks.

Programiz

Python Online Compiler

main.py

Visualize

Run ▶

```
1 #14
2 import numpy as np
3 np.random.seed(42)
4 marks = np.random.randint(0, 101, 5)
5 print("Random Marks:", marks)
6 print("Total Marks:", np.sum(marks))
7 print("Average Marks:", np.mean(marks))
8 print("Highest Marks:", np.max(marks))
9 print("Lowest Marks:", np.min(marks))
10 print("Standard Deviation:", round(np.std(marks), 2))
```

Output

Random Marks: [51 92 14 71 60]
Total Marks: 288
Average Marks: 57.6
Highest Marks: 92
Lowest Marks: 14
Standard Deviation: 25.74

Question 15 - Student Performance Analysis

The marks of five students in three subjects are stored in a NumPy array. Develop a program to perform a complete student performance analysis by calculating the total marks, average marks, highest marks, lowest marks, and identifying students who perform above the class

Average

main.py

Visualize

Run ▶

Output



```
1 import numpy as np
2 marks = np.array([
3     [78, 85, 90],
4     [65, 72, 80],
5     [88, 91, 84],
6     [56, 62, 70],
7     [95, 89, 92]
8 ])
9
10 total = np.sum(marks, axis=1)
11 average = np.mean(marks, axis=1)
12 highest = np.max(marks)
13 lowest = np.min(marks)
14 class_average = np.mean(marks)
15 above_average = np.where(
16     average > class_average
17 )[0] + 1
18 print("Student Marks:")
19 print(marks)
20 print("Student-wise Total:", total)
21 print("Student-wise Average:", np.round(average, 2))
22 print("Highest Marks:", highest)
23 print("Lowest Marks:", lowest)
24 print("Class Average:", round(class_average, 2))
25 print("Students Above Class Average:", above_average)
```

```
Student Marks:
[[78 85 90]
 [65 72 80]
 [88 91 84]
 [56 62 70]
 [95 89 92]]
Student-wise Total: [253 217 263 188 276]
Student-wise Average: [84.33 72.33 87.67 62.67 92. ]
Highest Marks: 95
Lowest Marks: 56
Class Average: 79.8
Students Above Class Average: [1 3 5]
```